

Jacobians Part 2

EGR 445/545

GVSU

Plan for Today

- Jacobian review
- Two link Jacobian
- Three link with twist
- ~~Three link planar~~

LA 12

Conceptual Review

- what is a Jacobian?
- what is a singularity?
 - mathematically
 - physically

Matrix of partial derivatives in Robotics:

$$\begin{bmatrix} \overline{J_{T,P}} \\ \overline{\omega_{linkN}} \end{bmatrix} = [J] \begin{bmatrix} \dot{\theta} \end{bmatrix}$$

tip Velocity
and rotational
velocity of last link

joint
Velocities

$\rightarrow \|J\| = 0$
 $\rightarrow [J]^{-1}$ does not exist

robot loses one
or more DoF

- I can no longer choose all my velocity components independently

How to Find the Jacobian

- find $\bar{v}_{tip}$ and $\bar{\omega}_N$ symbolically using `sympy` and velocity propagation
- take partial derivatives to find $\mathbf{J}$
- find singularity conditions based on $\|\mathbf{J}\| = 0$

eqn. 5.45 - 5.47

} `sympy`

Velocity Propagation

Conceptually:

- we work from the base out to the tip
- velocity of the end of a link is velocity at the base of the link plus the rotational part:

5.46

$$\vec{v}_{tip} = \vec{v}_{base} + \vec{\omega} \times \vec{r}$$

- ~~$\vec{\omega}$ is the total rotational velocity of the link, including rotation of all lower numbered joints~~
- $\vec{r}$ is the position vector of the tip with respect to the base of the link
- vectors for cross products must be in the same frame

velocity of tip w.r.t. base
caused by rotation

Velocity Propagation Equations

$(0 \hat{e}_{i+1}) \rightarrow \downarrow + v_p$
 $\swarrow \text{base } (0 \hat{i})$
 • eqn. 5.46: ${}^i v_{i+1} = {}^i v_i + {}^i \bar{\omega}_i \times {}^i P_{i+1}$ ← $\bar{\omega} \times \bar{r}$

• example: ${}^1 v_2 = {}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2$

• eqn. 5.47K: ${}^{i+1} v_{i+1} = {}_i^{i+1} \mathbf{R} \cdot {}^i v_{i+1}$

• example: ${}^2 v_2 = {}_1^2 \mathbf{R} ({}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2) = {}_1^2 \mathbf{R} {}^1 v_2$

• eqn. 5.45: ${}^{i+1} \bar{\omega}_{i+1} = {}_i^{i+1} \mathbf{R} {}^i \bar{\omega}_i + \dot{\theta}_{i+1} {}^{i+1} \hat{Z}_{i+1}$

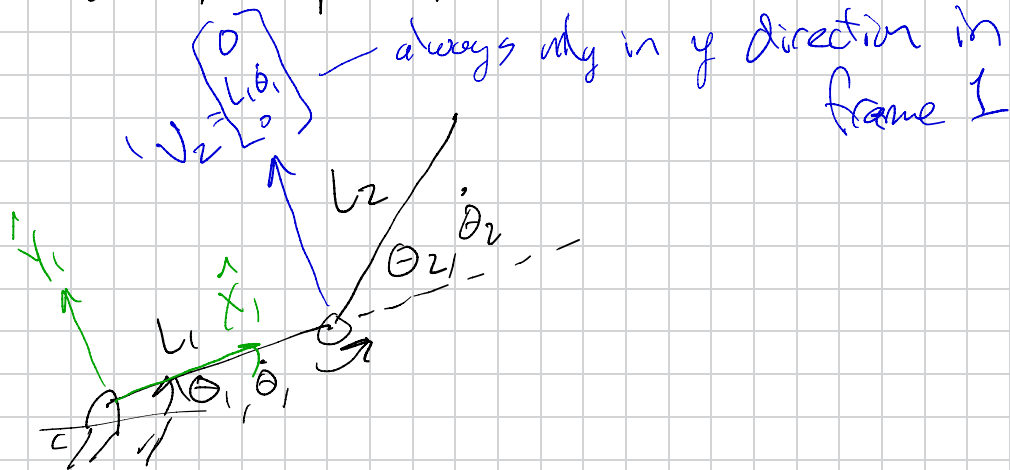
${}^2 R$ or ${}^3 R$
 ${}^2 F$

Two-Link Planar Jacobian

- two link with arbitrary angles
- how do we find the singularity condition(s)?
- walk through all the steps using `sympy`

`two_link_Jacobian_full_example.ipynb`

Two Link planar Jacobian



$${}^i \bar{V}_{i+1} = {}^i \bar{V}_i + {}^i \bar{\omega}_i \times {}^i \bar{P}_{i+1} \quad \} \text{eqn. 5.46}$$

$${}^{i+1} \bar{V}_{i+1} = {}^{i+1} R^i {}^i \bar{V}_{i+1} \quad \} \text{eqn. 5.47 alt. 5.46}$$

Proof 5.47:

$${}^{i+1} \bar{V}_{i+1} = {}^{i+1} R^i \left({}^i \bar{V}_i + {}^i \bar{\omega}_i \times {}^i \bar{P}_{i+1} \right)$$

Two Link planar

Jacobian

$${}^2\bar{V}_2 = {}^2P_1' \bar{V}_1$$

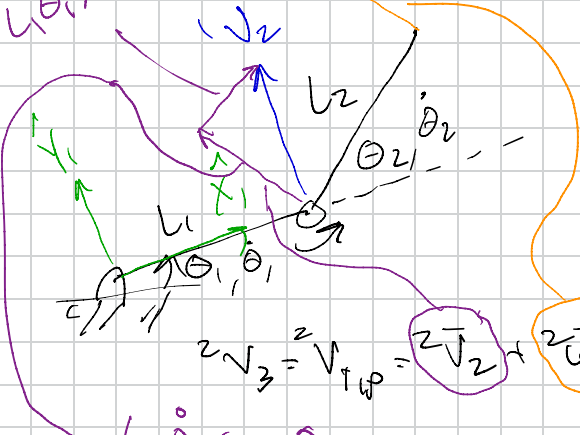
$${}^2\bar{V}_2 = {}^2P_1' \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix}$$

$${}^2\bar{P}_3 = \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix} {}^1\bar{\omega}_1$$

$${}^2\bar{V}_3 = {}^2V_{tip} = {}^2\bar{V}_2 + {}^2\bar{\omega}_2 \times {}^2\bar{P}_3$$

$$L_1 \dot{\theta}_1 \sin \theta_2$$

$$L_2(\dot{\theta}_1 + \dot{\theta}_2)$$

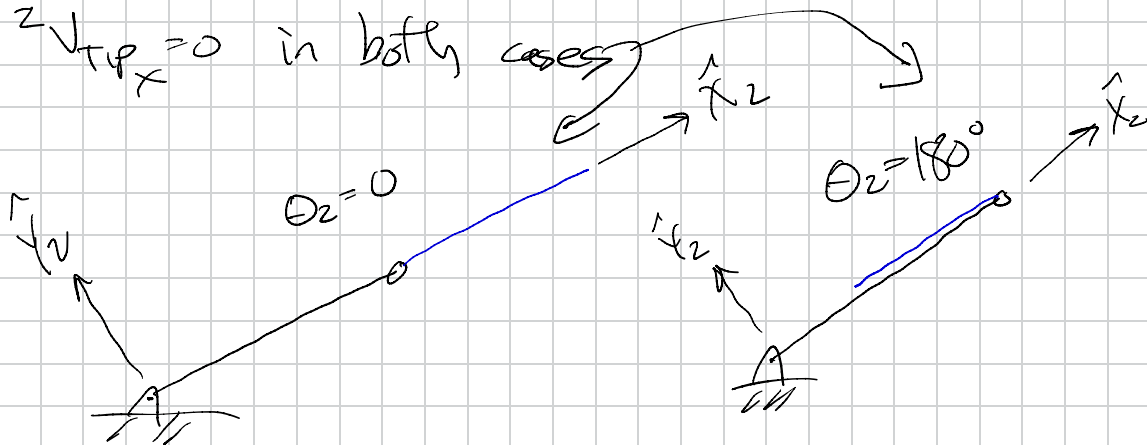


$$L_1 \dot{\theta}_1 \cos \theta_2$$

Two Link Singularity Condition

$$\sin \theta_2 = 0$$

${}^zV_{TPX} = 0$ in both cases



Three Link with Twist

- add third link to two-link planar, but $\alpha_2 = 90^\circ$

`three_link_with_twist_Jacobian_example.ipynb`

Jacobian Learning Activity

- LA 12